Scatter-Regularized Cross-Multi-Kernel Contrastive Learning for One-Class Anomaly Detection

, , , ,

Abstract

We present the methodology of a scatter-regularized cross-multi-kernel contrastive framework (SCMK) for one-class anomaly detection. The full manuscript is in preparation; this file contains the Methodology section.

Keywords: anomaly detection, multiple kernel learning, contrastive learning, one-class classification, representation learning

1. The proposed SCMK method

1.1. Problem formulation and notation

Let $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^{N}$ with $\boldsymbol{x}_i \in \mathbb{R}^D$ denote the data after preprocessing,[1] and let $y_i \in \{0, 1\}$ be the latent label, where $y_i = 0$ marks a normal instance and $y_i = 1$ an anomaly. Only the normal subset $\mathcal{X}_\mathrm{n} = \{\boldsymbol{x}_i : y_i = 0\}$, of size $N_\mathrm{n} = |\mathcal{X}_\mathrm{n}|$, is available at training time; anomalies are neither observed nor simulated. The goal is to learn a scoring function $a : \mathbb{R}^D \to \mathbb{R}$ such that $a(\boldsymbol{x})$ is large for anomalies and small for normal points.

Our framework, termed *scatter-regularized cross-multi-kernel* contrastive learning (SCMK), couples a multiple-kernel contrastive encoder with a one-class detection head. It comprises four components, summarised in Algorithm 1: (i) a multi-scale kernel bank that injects geometric priors at several resolutions (Section 1.2); (ii) a cross-kernel contrastive objective that aligns the views produced by different kernels (Section 1.3); (iii) a multi-kernel scatter regularizer that explicitly compacts the normal class in every kernel-induced embedding (Section 1.4); and (iv) a dual-signal one-class detector

[1] Nominal attributes are one-hot encoded and numerical attributes are min–max scaled to $[0, 1]$, so that all feature dimensions share a comparable range.

that fuses an angular and a radial OC-SVM score to flag anomalies (Section 1.6).

1.2. Multi-scale kernel bank

To endow the encoder with multi-resolution sensitivity, we instantiate a bank of K Gaussian kernels whose bandwidths are anchored to the intrinsic scale of the normal data. Let

$$\mathrm{med} = \mathrm{median}\big\{ \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 : \boldsymbol{x}_i, \boldsymbol{x}_j \in \mathcal{X}_\mathrm{n},\ i \neq j \big\} \tag{1}$$

be the median pairwise Euclidean distance estimated *from normal samples only*, which prevents contamination of the scale statistic by outliers. The k-th kernel is

$$\kappa_k(\boldsymbol{a}, \boldsymbol{b}) = \exp\left(-\frac{\|\boldsymbol{a} - \boldsymbol{b}\|_2^2}{2\, t_k^2} \right), \qquad t_k = r_k \cdot \mathrm{med}, \tag{2}$$

with multiplicative ratios $\{r_k\}_{k=1}^K = \{0.1, 0.5, 1.0, 2.0, 5.0\}$ ($K = 5$). Small ratios produce sharp kernels that resolve fine-grained local neighbourhoods, whereas large ratios produce broad kernels that capture the global manifold geometry. The bank therefore provides K complementary similarity measures that the contrastive stage will reconcile.

1.3. Cross-kernel contrastive representation learning

Per-kernel embeddings.. For each kernel we attach an independent, bias-free linear projection head $g_k(\boldsymbol{x}) = \boldsymbol{W}_k \boldsymbol{x}$ with $\boldsymbol{W}_k \in \mathbb{R}^{d \times D}$, followed by ℓ_2 normalization,

$$\boldsymbol{h}_{k,i} = \frac{\boldsymbol{W}_k \boldsymbol{x}_i}{\|\boldsymbol{W}_k \boldsymbol{x}_i\|_2} \in \mathbb{S}^{d-1}, \qquad k = 1, \ldots, K, \tag{3}$$

so that every embedding lives on the unit hypersphere $\mathbb{S}^{d-1}$. Removing the bias keeps the representation centred at the origin, which is consistent with the origin-separating hyperplane assumption of the downstream OC-SVM, while the normalization equalises the scale across kernels.

Cross-kernel positive pairs.. Whereas conventional contrastive learning forms positive pairs through stochastic data augmentation, we treat the *kernel index as the view*: the two embeddings $\boldsymbol{h}_{k,i}$ and $\boldsymbol{h}_{l,i}$ of the *same* sample under *different* kernels constitute a positive pair, and embeddings of distinct samples are negatives. Consider a mini-batch of B normal samples and a kernel pair (k, l). Stacking the two views yields $\boldsymbol{F}^{(kl)} = [\boldsymbol{h}_{k,1}, \ldots, \boldsymbol{h}_{k,B}, \boldsymbol{h}_{l,1}, \ldots, \boldsymbol{h}_{l,B}]^\top \in \mathbb{R}^{2B\times d}$, whose rows are indexed by $m, n \in \{1, \ldots, 2B\}$. The pairwise affinity is the *symmetrised kernel response* evaluated on the embeddings,

$$s_{mn}^{(kl)} = \tfrac{1}{2}\Big[\kappa_k(\boldsymbol{f}_m, \boldsymbol{f}_n) + \kappa_l(\boldsymbol{f}_m, \boldsymbol{f}_n)\Big]. \tag{4}$$

Because the embeddings are unit-norm, the Gaussian response reduces to a monotone function of cosine similarity, $\kappa_k(\boldsymbol{f}_m, \boldsymbol{f}_n) = \exp\big(-(1-\langle \boldsymbol{f}_m,\ \boldsymbol{f}_n\rangle)/t_k^2\big)$, so t_k^2 plays the role of a contrastive temperature that is automatically adapted to the data scale through Eq. (1).

Cross-kernel InfoNCE.. For anchor n, let $\pi(n)$ index its cross-view positive (i.e. $\pi(n) = n + B$ for $n \leq B$ and $\pi(n) = n - B$ otherwise). The InfoNCE loss for the pair (k, l) is

$$\mathcal{L}_{\text{con}}^{(kl)} = -\frac{1}{2B}\sum_{n=1}^{2B} \log \frac{\exp\big(s_{n,\pi(n)}^{(kl)}\big)}{\sum\limits_{m\neq n} \exp\big(s_{nm}^{(kl)}\big)}, \tag{5}$$

where the denominator excludes the self-similarity term $m = n$. Averaging over all $\binom{K}{2}$ unordered kernel pairs gives the cross-kernel contrastive objective

$$\mathcal{L}_{\text{con}} = \frac{1}{\binom{K}{2}} \sum_{1\leq k<l\leq K} \mathcal{L}_{\text{con}}^{(kl)}. \tag{6}$$

Minimising Eq. (6) drives the encoder to a *view-invariant* representation in which all kernels agree on the embedding of each normal sample, i.e. $\boldsymbol{h}_{k,i} \approx \boldsymbol{h}_{l,i}$, while keeping distinct samples mutually discriminable through the negative terms.

1.4. Multi-kernel scatter regularization

Motivation.. The contrastive objective (6) enforces *cross-kernel consistency* but does not constrain the *absolute compactness* of the normal class: it is invariant to a rotation that spreads normal embeddings across the sphere as long as the per-sample views remain aligned. A one-class detector, however, benefits directly from a tightly concentrated normal region. We therefore introduce a regularizer that minimises the dispersion of the normal class within every kernel-induced embedding space, drawing on the one-class specialisation of centroid multiple-kernel k-means (CMKKM).

Definition.. Let $\boldsymbol{\mu}_k = \frac{1}{B}\sum_{i=1}^{B} \boldsymbol{h}_{k,i}$ be the empirical centroid of the normal embeddings under kernel k. The multi-kernel scatter penalty is

$$\mathcal{L}_{\text{scat}} = -\frac{1}{K}\sum_{k=1}^{K} \|\boldsymbol{\mu}_k\|_2^2 . \tag{7}$$

It is computed in $O(KBd)$ time — a single mean per kernel — and thus adds negligible overhead.

Exact equivalence to within-class scatter.. The next result shows that Eq. (7) is not a heuristic surrogate but is *exactly* (up to an additive constant) the average within-class scatter of the normal embeddings.

Proposition 1 (Scatter equivalence)**.** *For unit-norm embeddings* $\boldsymbol{h}_{k,i} \in \mathbb{S}^{d-1}$, *define the average normalised within-class scatter* $\sigma^2 = \frac{1}{NK}\sum_{k=1}^{K}\sum_{i=1}^{N} \|\boldsymbol{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2$. *Then*

$$\sigma^2 = 1 + \mathcal{L}_{\text{scat}}. \tag{8}$$

Consequently, minimising $\mathcal{L}_{\text{scat}}$ *is equivalent to minimising the within-class scatter* σ^2, *and equivalently to maximising the mean intra-class kernel alignment* $\frac{1}{NK}\sum_k \sum_{i,j} \langle \boldsymbol{h}_{k,i},\, \boldsymbol{h}_{k,j}\rangle = \frac{1}{K}\sum_k \|\boldsymbol{\mu}_k\|_2^2$.

Proof. Fix a kernel k. Expanding the squared norm and using $\|\boldsymbol{h}_{k,i}\|_2^2 = 1$,

$$\begin{aligned}\sum_{i=1}^{N} \|\boldsymbol{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2 &= \sum_{i=1}^{N}\Big(\|\boldsymbol{h}_{k,i}\|_2^2 - 2\langle \boldsymbol{h}_{k,i},\, \boldsymbol{\mu}_k\rangle + \|\boldsymbol{\mu}_k\|_2^2 \Big) \\ &= N - 2N\|\boldsymbol{\mu}_k\|_2^2 + N\|\boldsymbol{\mu}_k\|_2^2 = N\big(1 - \|\boldsymbol{\mu}_k\|_2^2\big), \end{aligned} \tag{9}$$

where $\sum_i \langle \boldsymbol{h}_{k,i},\, \boldsymbol{\mu}_k\rangle = N\langle \boldsymbol{\mu}_k,\, \boldsymbol{\mu}_k\rangle = N\|\boldsymbol{\mu}_k\|_2^2$ follows from the definition of $\boldsymbol{\mu}_k$. Averaging Eq. (9) over k and dividing by N yields $\sigma^2 = 1 - \frac{1}{K}\sum_k \|\boldsymbol{\mu}_k\|_2^2 = 1 + \mathcal{L}_{\text{scat}}$. The alignment identity is immediate from $\sum_{i,j}\langle \boldsymbol{h}_{k,i},\, \boldsymbol{h}_{k,j}\rangle = N^2\|\boldsymbol{\mu}_k\|_2^2$. □

Remark 1 (Connection to one-class CMKKM)**.** *The one-class instance of centroid multiple-kernel k-means maximises* $\mathrm{Tr}(\boldsymbol{H}^\top \boldsymbol{K}_\eta \boldsymbol{H})$ *with a single soft cluster indicator* $\boldsymbol{H} = N^{-1/2}\mathbf{1}$ *and a convex kernel combination* $\boldsymbol{K}_\eta = \sum_k \eta_k\, \Phi_k \Phi_k^\top$, *where* $\Phi_k = [\boldsymbol{h}_{k,1}, \ldots, \boldsymbol{h}_{k,N}]^\top$ *realises a linear kernel on the embeddings. A short calculation gives* $\mathrm{Tr}(\boldsymbol{H}^\top \boldsymbol{K}_\eta \boldsymbol{H}) = N\sum_k \eta_k \|\boldsymbol{\mu}_k\|_2^2$, *which under uniform weights* $\eta_k = 1/K$ *equals* $-N\,\mathcal{L}_{\text{scat}}$. *Hence* $\mathcal{L}_{\text{scat}}$ *is precisely the (negated, scale-normalised) one-class CMKKM compactness objective, providing a multiple-kernel-learning justification for the penalty.*

1.5. Joint objective and collapse avoidance

The encoder $\{\boldsymbol{W}_k\}_{k=1}^{K}$ is trained by minimising the combined loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{con}} + \lambda\, \mathcal{L}_{\mathrm{scat}}, \tag{10}$$

where $\lambda \geq 0$ trades off cross-kernel consistency against intra-class compactness; $\lambda = 0$ recovers the purely contrastive backbone.

A natural concern is representational *collapse*: the scatter term alone is trivially minimised ($\mathcal{L}_{\mathrm{scat}} = -1$) when all embeddings of a kernel coincide ($\|\boldsymbol{\mu}_k\|_2 = 1$). This degenerate solution is, however, suppressed by the contrastive term. In the alignment–uniformity decomposition of contrastive learning, the numerator of Eq. (5) promotes *alignment* of positives while the log-sum-exp denominator promotes *uniformity* of the embedding distribution on the sphere, which is maximised only by a spread-out configuration and is therefore directly opposed to collapse. The joint objective (10) thus realises a controlled balance: $\mathcal{L}_{\mathrm{scat}}$ pulls each normal class towards its centroid, $\mathcal{L}_{\mathrm{con}}$ prevents the normal embeddings from degenerating to a point and keeps different samples separable, and λ governs the equilibrium. The unit-sphere constraint additionally bounds $\|\boldsymbol{\mu}_k\|_2 \leq 1$, so the penalty cannot diverge.

1.6. Dual-signal anomaly scoring

After training, the projection heads are frozen and each sample $\boldsymbol{x}_i$ is described by *two complementary signals* read off the raw projections $\boldsymbol{W}_k\boldsymbol{x}_i$: the *angular* position of its unit-norm embeddings and the *radial* magnitude of those projections. These signals carry non-redundant discriminative information, and which of them separates the normal and anomalous classes is dataset-dependent. We therefore build a detector for each and fuse their scores.

Directional signal: linear OC-SVM on the angular embedding.. The unit-norm embeddings $\boldsymbol{h}_{k,i}$ retain only the angular position of a sample on each kernel sphere—exactly the geometry that the scatter regularizer (7) compacts. Concatenating the K views,

$$\boldsymbol{z}_i = \left[\boldsymbol{h}_{1,i}^{\top}, \boldsymbol{h}_{2,i}^{\top}, \ldots, \boldsymbol{h}_{K,i}^{\top}\right]^{\top} \in \mathbb{R}^{Kd}, \tag{11}$$

fuses them into a single descriptor, and a linear-kernel OC-SVM fitted on the normal embeddings $\{\boldsymbol{z}_i : y_i = 0\}$ by solving

$$\min_{\boldsymbol{w},\rho,\boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|_2^2 - \rho + \frac{1}{\nu N_{\mathrm{n}}} \sum_{i:\, y_i=0} \xi_i \quad \text{s.t.} \quad \langle \boldsymbol{w},\, \boldsymbol{z}_i\rangle \geq \rho - \xi_i, \ \ \xi_i \geq 0, \tag{12}$$

yields the *directional score* $s_i^{\mathrm{dir}} = \rho - \langle \boldsymbol{w},\, \boldsymbol{z}_i \rangle$, where $\nu \in (0, 1]$ upper-bounds the fraction of normal training points allowed outside the decision region. A linear separator suffices here because the contrastive and scatter objectives jointly mould the normal class into a compact, nearly linearly separable cone on the product of spheres $(\mathbb{S}^{d-1})^K$; this score is informative whenever the normal/anomaly contrast is *angular*.

Magnitude signal: RBF OC-SVM on the projection norms.. The ℓ_2 normalization in Eq. (3) deliberately discards the projection norms $r_{k,i} = \|\boldsymbol{W}_k \boldsymbol{x}_i\|_2$. Yet these norms are themselves discriminative: trained on normal data alone, each head maps normal samples into a characteristic energy band, whereas anomalies—lying off the learned manifold—are projected to markedly different, typically larger, norms. We collect the per-kernel norms into a magnitude vector

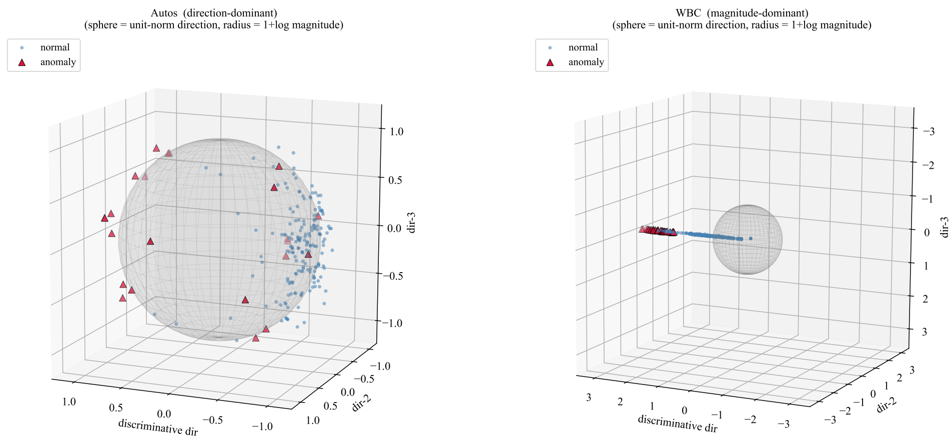
$$\boldsymbol{r}_i = \left[r_{1,i}, r_{2,i}, \ldots, r_{K,i}\right]^\top \in \mathbb{R}^K, \tag{13}$$

and fit an *RBF*-kernel OC-SVM on the normal norms $\{\boldsymbol{r}_i : y_i = 0\}$, producing the *magnitude score* s_i^{mag}. The RBF kernel is essential: a *linear* OC-SVM on norms admits a sign-reversed optimum—when the normal norms are small, the separating direction turns negative and labels large-norm anomalies as inliers—whereas the RBF density model, centred on the compact small-norm normal cluster, correctly assigns high scores to the large-norm outliers lying outside it. This score is informative whenever the contrast is *radial*.

Score fusion.. On a given dataset one signal may be essentially uninformative—most strikingly, the directional score collapses to near-chance when normal and anomalous samples are angularly indistinguishable but differ in projection energy. Without assuming *a priori* which regime holds, we fuse the two detectors by a per-sample maximum of min–max-normalised scores,

$$a(\boldsymbol{x}_i) = \max\Big(\widetilde{s}_i^{\mathrm{dir}},\ \widetilde{s}_i^{\mathrm{mag}}\Big), \qquad \widetilde{s}_i^{\bullet} = \frac{s_i^{\bullet} - \min_j s_j^{\bullet}}{\max_j s_j^{\bullet} - \min_j s_j^{\bullet}}, \tag{14}$$

so that a sample flagged by *either* detector receives a high anomaly score. The fusion needs no prior knowledge of which signal is informative and, as the ablation in Section **??** confirms, remains robust on datasets where either single signal collapses while never trailing the stronger signal by a meaningful margin. Fig. 1 makes this complementarity geometric: on a direction-dominant dataset the anomalies stay on the sphere but cluster to one angular region,



(a) Autos — direction-dominant

(b) WBC — magnitude-dominant

Figure 1: Geometric view of the two complementary signals on the learned representation. The grey sphere is the unit-norm (direction) manifold; a sample's radius is $1 + \log$ of its projection magnitude, so normal points (≈ 1) lie on the shell. **(a)** On Autos the anomalies remain on the shell but concentrate in one angular region—separable by *direction*, while their magnitude is uninformative. **(b)** On WBC the anomalies are angularly intermixed with the normal class yet bulge far outside the shell—separable only by *magnitude*. The angular (linear OC-SVM) detector handles (a), the radial (RBF OC-SVM) detector handles (b), and the max-fusion is robust on both without knowing the regime in advance. For visualization only, the principal in-sphere axis is the normal/anomaly mean-difference direction (labels are used solely for this projection, never for training or scoring).

whereas on a magnitude-dominant dataset they are angularly mixed with the normal class yet bulge far outside the sphere—so a single detector necessarily fails on one regime, while the max-fusion captures both.

1.7. Overall algorithm and complexity

Algorithm 1 summarises the complete training and scoring pipeline. The dominant per-mini-batch cost is the cross-kernel contrastive term, $O\big(K^2B^2d\big)$ for the $\binom{K}{2}$ pairwise affinity matrices, while the scatter regularizer contributes only $O(KBd)$. Embedding extraction over the full data set is $O(NKdD)$. Scoring fits two OC-SVMs on the normal set: a linear one on the Kd-dimensional embeddings and an RBF one on the K-dimensional magnitude vectors; both scale with the number of support vectors, and the radial detector is negligible since $K = 5$. With a fixed small kernel bank, the method retains the same asymptotic complexity as the contrastive backbone, so the

Algorithm 1 SCMK for one-class anomaly detection

Require: normal set $\mathcal{X}_\mathrm{n}$, full set $\mathcal{X}$, kernel ratios $\{r_k\}_{k=1}^{K}$, embedding dim. d, weight λ, batch size B, epochs T, OC-SVM parameter ν

Ensure: anomaly scores $\{a(\boldsymbol{x}_i)\}_{i=1}^{N}$

med $\leftarrow$ median pairwise distance on $\mathcal{X}_\mathrm{n}$ ▷ Eq. (1)

build kernels κ_k with $t_k = r_k\,\mathrm{med}$ ▷ Eq. (2)

initialise projection heads $\{\boldsymbol{W}_k\}_{k=1}^{K}$

for epoch $= 1$ **to** T **do**

    for each mini-batch of B samples from $\mathcal{X}_\mathrm{n}$ **do**

        $\boldsymbol{h}_{k,i} \leftarrow \boldsymbol{W}_k\boldsymbol{x}_i / \left\|\boldsymbol{W}_k\boldsymbol{x}_i\right\|_2$ for all k, i ▷ Eq. (3)

        $\mathcal{L}_\mathrm{con} \leftarrow$ cross-kernel InfoNCE over kernel pairs ▷ Eqs. (4)–(6)

        $\mathcal{L}_\mathrm{scat} \leftarrow -\frac{1}{K}\sum_k \left\|\frac{1}{B}\sum_i \boldsymbol{h}_{k,i}\right\|_2^2$ ▷ Eq. (7)

        update $\{\boldsymbol{W}_k\}$ by a gradient step on $\mathcal{L}_\mathrm{con} + \lambda\mathcal{L}_\mathrm{scat}$ ▷ Eq. (10)

    end for

end for

$\boldsymbol{z}_i \leftarrow [\boldsymbol{h}_{1,i}; \ldots; \boldsymbol{h}_{K,i}]$, $\boldsymbol{r}_i \leftarrow [\|\boldsymbol{W}_1\boldsymbol{x}_i\|; \ldots; \|\boldsymbol{W}_K\boldsymbol{x}_i\|]$ for all $\boldsymbol{x}_i \in \mathcal{X}$ ▷ Eqs. (11),(13)

$s^\mathrm{dir} \leftarrow$ linear OC-SVM on $\{\boldsymbol{z}_i : y_i = 0\}$; $s^\mathrm{mag} \leftarrow$ RBF OC-SVM on $\{\boldsymbol{r}_i : y_i = 0\}$ ▷ Eq. (12)

return $a(\boldsymbol{x}_i) = \max\left(\widetilde{s}_i^\mathrm{dir}, \widetilde{s}_i^\mathrm{mag}\right)$ ▷ Eq. (14)

accuracy gains reported in the experiments come at no additional order-of-growth cost.

References

[1] A. van den Oord, Y. Li, O. Vinyals, *Representation learning with contrastive predictive coding*, arXiv:1807.03748, 2018.

[2] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, *A simple framework for contrastive learning of visual representations*, in: ICML, 2020, pp. 1597–1607.

[3] T. Wang, P. Isola, *Understanding contrastive representation learning through alignment and uniformity on the hypersphere*, in: ICML, 2020, pp. 9929–9939.

[4] M. Gönen, E. Alpaydın, *Multiple kernel learning algorithms*, Journal of Machine Learning Research 12 (2011) 2211–2268.

[5] H.-C. Huang, Y.-Y. Chuang, C.-S. Chen, *Multiple kernel fuzzy clustering*, IEEE Transactions on Fuzzy Systems 20 (1) (2012) 120–134.

[6] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, R. C. Williamson, *Estimating the support of a high-dimensional distribution*, Neural Computation 13 (7) (2001) 1443–1471.

[7] L. Ruff, R. Vandermeulen, N. Görnitz, et al., *Deep one-class classification*, in: ICML, 2018, pp. 4393–4402.